Ananth Ram Tekkalakota

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SUMMARY

CS student at UT Dallas (minor in Finance) graduating December 2026, who recently led backend architecture and built a GPT-powered natural-language-to-SQL engine for an enterprise claims analytics platform. Currently shipping an iOS app that uses an AI vision pipeline to transcribe handwritten music notation, now in TestFlight beta with external testers. Comfortable across the stack in TypeScript, Python, and Next.js, with ML work spanning classical models and LSTMs.

EDUCATION

University of Texas at Dallas — Aug 2022 - Dec 2026

Bachelor of Science, Computer Science, Minor in Finance

- **Coursework:** Data Structures and Algorithms, System Programming and UNIX, Computer Architecture, Software Engineering, Probability and Statistics, Artificial Intelligence, Advanced Algorithms Design and Analysis, Intro to Machine Learning

WORK EXPERIENCE

Smart Data Solutions | *Backend Lead & AI Developer* — Aug 2025 - Dec 2025

- Led backend architecture for an enterprise claims analytics platform using Next.js, Prisma ORM, and MySQL, designing and optimizing schemas for users, claims, queues, and widget preferences to support complex reporting.
- Built an AI-powered natural language–to–SQL query engine using GPT models, including LLM integration, a SQL validation/safety layer, and RESTful API routes so non-technical users can run secure ad-hoc queries.
- Implemented NextAuth.js authentication with bcrypt-hashed credentials, JWT sessions, and role-based access control for admins and analysts.
- Developed an admin data management console with CRUD operations and CSV import for claims data, streamlining dataset onboarding and quality control.
- Created a customizable dashboard and widget system (including chart widgets) and collaborated with the frontend team to integrate APIs, improving usability and data visualization.

Spark My Sport | *Full Stack Developer Intern* — Jan 2024 - Mar 2024

- Developed a Typescript program to convert spreadsheet data into JSON files, utilizing object-oriented programming techniques to enhance data interoperability and accessibility.
- Engineered a system using Next.js and JavaScript to visually display JSON data, facilitating real-time data interpretation and decision-making.
- Optimized APIs to pull data and videos from the backend, improving search algorithm efficiency by 15% while documenting code for maintainability and future reference.

AIS | *Team Member* — Feb 2023 - Apr 2023

- Developed a food-scanning AI system using Python's OpenCV library, improving food recognition accuracy by 20%
- Devised strategic solutions to optimize the local OpenCV model, reducing processing time by 10%

PROJECTS

CarnaticTranslate - iOS App (Swift, SwiftUI) https://testflight.apple.com/join/XdHuceH9 — Jun 2026 - Present

- Designed and shipped a solo iOS app to TestFlight that photographs handwritten Carnatic music sheets and returns a structured, editable transcription; currently in beta with external testers
- Engineered an AI vision pipeline (Gemini and OpenAI models) extracting swaras, sahitya, ragam, and talam into a typed schema, with retry logic and truncation handling to make unreliable model output production-safe
- Implemented iCloud sync via a CloudKit ubiquity container for automatic multi-device backup, plus a versioned-envelope JSON export format with a custom UTI for sharing transcriptions across devices
- Built paginated PDF export with content-proportional column widths and shipped iterative features — cell copy/paste/move, multi-song detection in multi-page imports — driven directly by beta tester feedback

Sentiment Analysis Classifier https://github.com/dp019/4375-project — May 2025

- Co-developed a sentiment classification pipeline benchmarking Naive Bayes, Linear SVM, and Logistic Regression across three domain datasets (Amazon, IMDB, Yelp), totaling 100K+ reviews with 85%+ test accuracy
- Engineered per-domain preprocessing (HTML stripping, stopword filtering, rating-to-label conversion) and TF-IDF matrices capped at 5,000 features, improving baseline accuracy by ~10% through vectorizer tuning
- Implemented a stacked LSTM (64→32 units) in TensorFlow/Keras with dropout and early stopping, boosting accuracy ~5% over the best traditional model by capturing sequential context
- Compared all models on precision, recall, and F1 via classification reports to select the best algorithm per dataset

Skills

- **Languages:** Python, Java, C++, TypeScript, JavaScript, SQL, HTML/CSS
- **Frameworks & Libraries:** Next.js, React, Flask, Flutter, TensorFlow/Keras, Scikit-learn, OpenCV
- **Tools & Platforms:** Git/GitHub, Prisma, MySQL, Restful API, OpenAI API, RAG, Agile, OOP, DSA, AWS (S3, CloudFront, IAM)